PROJECT / ASSIGNMENT – 1

Economic Selection of Two Heterogeneous Generating Plants

Group Number:	11
Member Number:	4
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EE315 – Power Systems

1. Objective

To develop a generalized program for the economic selection of two heterogeneous generating plants, identify the base-load and peak-load plants, and determine the operating requirement of the peak-load plant for the assigned Group 11, Member 4 data.

2. Assignment Data Generation

For Group Number = 11 and Member Number = 4:

Table 1: Generation of assigned numerical values

Parameter	Rule	Calculation	Result
XX	Group Number	11	11
ZZ	$\lvert$Group Number $-$ 10$\rvert$	$\lvert 11 - 10 \rvert$	1
Y	Member Number	4	4
K	$\lvert$9 $-$ Member Number$\rvert$	$\lvert 9 - 4 \rvert$	5
J	$\lvert$17 $-$ Member Number$\rvert$	$\lvert 17 - 4 \rvert$	13
Maximum demand	5XX MW	511	511 MW
Load factor	5Y %	54	54%
Plant 1 fixed cost	YXX Rs/kW/year	411	Rs 411/kW/year
Plant 1 running cost	Y p/kWh	4	4 p/kWh
Plant 2 fixed cost	KZZ Rs/kW/year	501	Rs 501/kW/year
Plant 2 running cost	J p/kWh	13	13 p/kWh

3. Theory

If a plant supplies 1 kW of capacity for H hours in a year, its annual cost per kW is the sum of annual fixed cost and running energy cost.

For Plant 1,

$$TC_1 = C_1 + \frac{R_1}{100} H$$

For Plant 2,

$$TC_2 = C_2 + \frac{R_2}{100} H$$

where C_1, C_2 are in Rs/kW/year and R_1, R_2 are in paise/kWh.

A normal base-load/peak-load crossover exists only when one plant has the higher fixed cost but lower running cost, while the other has the lower fixed cost but higher running cost.

If one plant has both lower fixed cost and lower running cost, that plant economically dominates the other plant for every non-negative operating duration. In that case, the dominated plant is not dispatched in the pure minimum-cost solution.

4. Calculations

4.1 Average Demand

Average Demand is given by

$$\text{Average Demand} = \text{Maximum Demand} \times \text{Load Factor}$$

$$= 511 \times 0.54$$

$$\boxed{\text{Average Demand} = 275.94 \text{ MW}}$$

4.2 Annual Energy Requirement

The annual energy requirement is

$$\text{Annual Energy} = \text{Maximum Demand} \times \text{Load Factor} \times 8760$$

$$= 511 \times 0.54 \times 8760$$

$$\boxed{\text{Annual Energy} = 2,417,234.40 \text{ MWh/year}}$$

$$= 2,417.2344 \text{ GWh/year}$$

4.3 Comparison of the Two Plants

Table 2: Comparison of the two generating plants

Cost Component	**Plant 1**	**Plant 2**
Fixed Cost	Rs 411/kW/year	Rs 501/kW/year
Running Cost	4 p/kWh	13 p/kWh

Plant 1 has a lower fixed cost,

$$411 < 501$$

and a lower running cost,

$$4 < 13.$$

Therefore, Plant 1 is cheaper than Plant 2 at all positive operating hours.

4.4 Crossover Check

Set

$$TC_1 = TC_2$$

Therefore,

$$411 + \frac{4}{100}H = 501 + \frac{13}{100}H$$

$$411 + 0.04H = 501 + 0.13H$$

$$411 - 501 = 0.13H - 0.04H$$

$$-90 = 0.09H$$

Hence,

$$\boxed{H = -1000 \text{ hours/year}}$$

The result is -1000 hours/year, which is not a feasible operating duration because yearly operating hours cannot be negative. The negative crossover indicates that the two cost curves do not intersect for any physically feasible value of H.

Since Plant 1 has both the lower fixed cost and lower running cost, it remains cheaper than Plant 2 throughout the feasible operating range.

5. Base-Load and Peak-Load Decision

Table 3: Base-load and peak-load decision

Decision	Result
Base-load plant	Plant 1
Peak-load plant	Plant 2, but not economically dispatched
Feasible positive crossover	None
Peak-load plant operating hours	0 hours/year
Peak-load plant operating power	0 MW under the minimum-cost solution

Thus, for the assigned numerical values, the minimum-cost generating system would use Plant 1 throughout the required operating range. Plant 2 would have zero economic operating hours.

6. Cost-Curve Verification

At $H = 0$:

$$TC_1 = 411 \text{ Rs/kW-year}$$

and

$$TC_2 = 501 \text{ Rs/kW-year}.$$

At $H = 8760$:

$$TC_1 = 411 + (0.04)(8760)$$

$$= 411 + 350.40$$

$$\boxed{TC_1 = 761.40 \text{ Rs/kW-year}}$$

For Plant 2:

$$TC_2 = 501 + (0.13)(8760)$$

$$= 501 + 1138.80$$

$$\boxed{TC_2 = 1639.80 \text{ Rs/kW-year}}$$

Plant 1 remains cheaper at both ends of the feasible operating interval and, because it also has the lower running-cost slope, it remains cheaper for every H between 0 and 8760 hours.

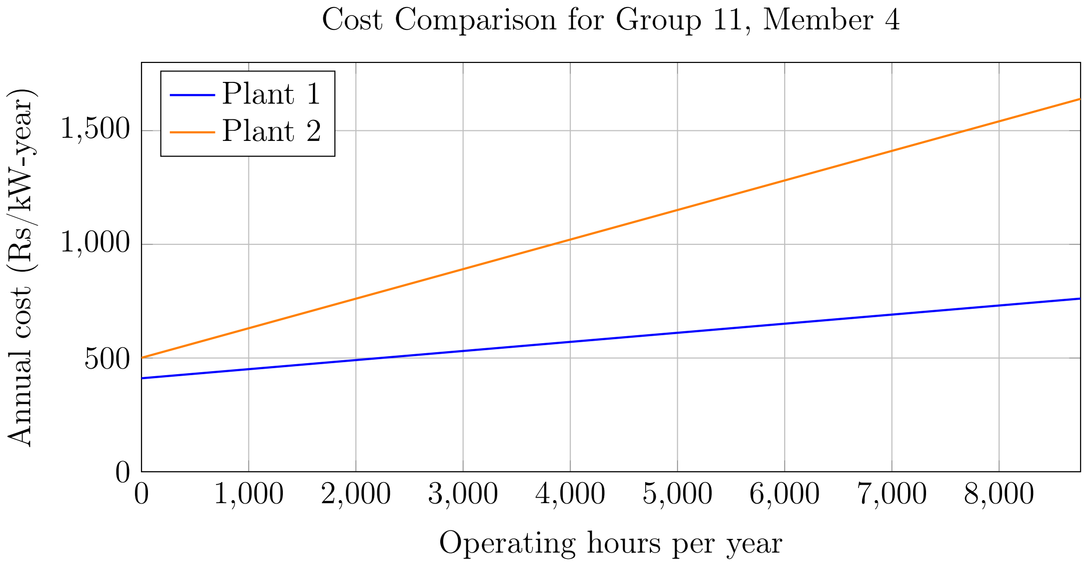

Figure 1: Annual cost curves for Group 11, Member 4.

7. Algorithm

1. Read P, LF, C_1, R_1, C_2 and R_2.
2. Calculate annual energy requirement.
3. Check whether Plant 1 is lower in both fixed and running cost.
4. Check whether Plant 2 is lower in both fixed and running cost.
5. If one plant dominates, select it as the economic base plant and assign 0 operating hours to the economically dominated plant.
6. Only if a fixed-cost/running-cost trade-off exists, solve the signed crossover equation.
7. Reject negative or out-of-range crossover values instead of using an absolute value.
8. Display the selected plants and operating result.

8. MATLAB Program

The submitted MATLAB program is generalized. It accepts the six problem inputs and automatically handles both normal crossover cases and dominance cases such as Group 11, Member 4.

```

clc;
clear;

% -----------------------------------------------------------
% Assignment 1 - Economic Selection of Two Generating Plants
% Correct generalized program
% Group Number = 11, Member Number = 4
% -----------------------------------------------------------

% Input

P = input('Enter Maximum Demand (MW): ');
LF = input('Enter Load Factor (%): ');

C1 = input('Enter Fixed Cost of Plant 1 (Rs/kW/year): ');
R1 = input('Enter Running Cost of Plant 1 (p/kWh): ');

C2 = input('Enter Fixed Cost of Plant 2 (Rs/kW/year): ');
R2 = input('Enter Running Cost of Plant 2 (p/kWh): ');


% Annual energy requirement

energy_MWh = P * 8760 * (LF/100);

fprintf('\nAnnual Energy Requirement = %.2f MWh/year\n', energy_MWh);


% Annual cost equations:
%
% TC1 = C1 + (R1/100)*H
% TC2 = C2 + (R2/100)*H


% Check whether Plant 1 economically dominates Plant 2

if (C1 <= C2 && R1 <= R2) && (C1 < C2 || R1 < R2)

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```
    base = "Plant 1";
    peak = "Plant 2";
    peakHours = 0;

    fprintf('\nPlant 1 has both lower fixed cost and lower/equal running cost.\n');
    fprintf('Therefore Plant 1 economically dominates Plant 2.\n');

    fprintf('Base Load Plant = %s\n', base);
    fprintf('Peak Load Plant = %s (not economically dispatched)\n', peak);
    fprintf('Peak Load Plant Operating Hours = %.2f hours/year\n', peakHours);
    fprintf('Peak Load Plant Power = 0 MW under the minimum-cost solution.\n');


% Check whether Plant 2 economically dominates Plant 1

elseif (C2 <= C1 && R2 <= R1) && (C2 < C1 || R2 < R1)

    base = "Plant 2";
    peak = "Plant 1";
    peakHours = 0;

    fprintf('\nPlant 2 economically dominates Plant 1.\n');

    fprintf('Base Load Plant = %s\n', base);
    fprintf('Peak Load Plant = %s (not economically dispatched)\n', peak);
    fprintf('Peak Load Plant Operating Hours = %.2f hours/year\n', peakHours);
    fprintf('Peak Load Plant Power = 0 MW under the minimum-cost solution.\n');


% Normal fixed-cost/running-cost trade-off case

else

    if R2 ~= R1

        H = 100 * (C1 - C2) / (R2 - R1);

        fprintf('\nSigned crossover from cost equations = %.2f hours/year\n', H);

        if H >= 0 && H <= 8760

            if R1 < R2

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```
                base = "Plant 1";
                peak = "Plant 2";

            else

                base = "Plant 2";
                peak = "Plant 1";

            end

            peakHours = H;

            fprintf('Base Load Plant = %s\n', base);
            fprintf('Peak Load Plant = %s\n', peak);
            fprintf('Peak Load Plant Operating Hours = %.2f hours/year\n', peakHours);

        else

            fprintf('The crossover is outside the feasible range of 0 to 8760 hours/year.\n');
            fprintf('Therefore there is no feasible positive crossover.\n');

        end

    else

        fprintf('\nThe two plants have equal running costs.\n');

        if C1 < C2

            fprintf('Plant 1 is cheaper at all operating hours.\n');
            fprintf('Base Load Plant = Plant 1\n');
            fprintf('Peak Load Plant = Plant 2 (not economically dispatched)\n');
            fprintf('Peak Load Plant Operating Hours = 0 hours/year\n');

        elseif C2 < C1

            fprintf('Plant 2 is cheaper at all operating hours.\n');
            fprintf('Base Load Plant = Plant 2\n');
            fprintf('Peak Load Plant = Plant 1 (not economically dispatched)\n');
            fprintf('Peak Load Plant Operating Hours = 0 hours/year\n');

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        else

            fprintf('Both plants have identical costs.\n');

        end

    end

end
```

9. Expected Output for Group 11, Member 4

```
GROUP 11 - MEMBER 4

Inputs:

Maximum Demand P = 511 MW
Load Factor LF = 54 %
Plant 1: C1 = Rs 411/kW/year, R1 = 4 p/kWh
Plant 2: C2 = Rs 501/kW/year, R2 = 13 p/kWh

Annual Energy Requirement = 2417234.40 MWh/year
Average Demand = 275.94 MW

Plant 1 economically dominates Plant 2.
Base Load Plant = Plant 1
Peak Load Plant = Plant 2 (not economically dispatched)

Signed crossover = -1000.00 hours/year
This is negative, so no feasible positive crossover exists.

Peak Load Plant Operating Hours = 0 hours/year
Peak Load Plant Operating Power = 0 MW under the minimum-cost solution.
```

10. Final Answer

For Group 11, Member 4, Plant 1 is the economically preferred/base-load plant. Plant 2 is the nominal second/peak plant, but it is not economically dispatched because Plant 1 has both lower fixed and running costs.

The signed crossover is

$$\boxed{-1000 \text{ h/year}}$$

so no positive crossover exists.

Therefore, the peak-load plant operating hours are

$$\boxed{0 \text{ h/year}}$$

and its dispatched power is

$$\boxed{0 \text{ MW}}$$

under the minimum-cost solution.

11. Important Note

Maximum demand and load factor alone are not sufficient to determine a non-zero physical peak-plant operating duration in a normal two-plant case; a load-duration curve or equivalent demand-duration information would normally be required.

For the assigned Group 11, Member 4 data, however, Plant 1 has both a lower fixed cost and a lower running cost than Plant 2.

Therefore, Plant 1 economically dominates Plant 2, making the minimum-cost solution unambiguous: Plant 2 is not economically dispatched and has 0 operating hours/year.

12. GitHub Repository

https://github.com/K-SiriChandana/POWERSYSTEMASSIGNMENT1 **GitHub Repository – POWERSYSTEMASSIGNMENT1**